WEBSITE DOCUMENTATION

Farah Sayari

Portfolio content reference

06 November 2026 · Africa/Tunis

An exact content record of the local portfolio: every HTML page, its text, navigation, project entries, links, images, form fields, and dynamic text states.

Snapshot: 2026-11-06 · Africa/Tunis · 12 HTML pages. The source files are the authority for this record.

Document guide

Website wording is reproduced without rewriting. HTML entities are decoded and layout whitespace is normalized; line wrapping, bullets, and font treatment follow this document. Content follows HTML source order, so a two-column page is read column by column. Hover text and animated alternatives are recorded explicitly.

This is a content reference rather than a screenshot reproduction. Explanatory notes, inventories, and update instructions are separate from the sections marked “Exact page content.” External websites, the map service, and contact delivery were not verified; the contact form was not submitted.

Pages covered

No.	Page	Source file	Entry point
1	Home	index.html	Main navigation
2	About	about.html	Main navigation
3	Resume	resume.html	Main navigation
4	Certifications	certifications.html	Main navigation
5	Services	services.html	Main navigation
6	Portfolio	portfolio.html	Main navigation
7	LollyBooks project details	portfolio-details.html	Portfolio detail link
8	PulmoTrack project details	pulmotrack-details.html	Portfolio detail link
9	TestPilot project details	testpilot-details.html	Portfolio detail link
10	Digital Gallery	digital-gallery.html	Main navigation
11	Contact	contact.html	Main navigation
12	Starter template	starter-page.html	Retained template; absent from the main navigation

How to update this record

To update the website and its documentation: edit the relevant HTML, CSS, JavaScript, image, or CV file, then run the regeneration command below from the portfolio folder.

```
python documentation/generate.py
```

On Windows, you can also double-click documentation/Refresh Documentation.cmd. The script reads the current website and refreshes the PDF, editable Word file, HTML, Markdown, JSON inventory, and source snapshot archive.

For a documentation-only revision, open the .docx file in Word or LibreOffice and save/export it as PDF. This does not change the website. Regeneration replaces generated outputs, so preserve manual Word edits under a different filename.

Keep ongoing commentary or planned work in documentation/notes.json. Those notes survive regeneration and are labeled separately from current website content.

Where to make common changes

Change	Edit these sources
Home introduction / animated job titles	index.html; animation timing in assets/js/main.js
Biography / contact profile / rotating photos	about.html; assets/img/me11.png, me22.png, me33.png
Resume text / downloadable CV	resume.html; assets/docs/Farah_Sayari_CV2026.pdf
Service titles and descriptions	services.html
Digital drawings / gallery introduction	digital-gallery.html; assets/css/collections.css
Course certificates / credential links	certifications.html; assets/css/collections.css
Project list / filters / thumbnails	portfolio.html; assets/img/portfolio/ and assets/img/masonry-portfolio/
LollyBooks case study	portfolio-details.html; assets/img/portfolio/lollybooks/library-art.webp
TestPilot case study	testpilot-details.html; assets/css/portfolio.css
PulmoTrack case study / scientific paper	pulmotrack-details.html; assets/docs/PulmoTrackPaper.pdf; assets/img/portfolio/pulmotrack/paper-preview.png
Contact text / form fields / delivery endpoint	contact.html, including its inline JavaScript
Global header / footer	Repeated in each HTML file; update each affected page
Shared visual styles / page-specific styles	assets/css/main.css; assets/css/resume.css; assets/css/portfolio.css; assets/css/collections.css
Shared navigation / animations / filtering	assets/js/main.js

Shared header, navigation, and footer

Use the table below with the exact variants that follow. Each page's visible brand, navigation labels, social destinations, footer text, and active navigation item are included.

Page	Brand	Navigation	Header social	Footer	Active item
index.html	FARAH	N1	S1	F1	Home
about.html	FARAH	N1	S1	F1	About
resume.html	FARAH	N1	S1	F2	Resume
certifications.html	FARAH	N1	S1	F1	Certifications
services.html	FARAH	N1	S1	F1	Services
portfolio.html	FARAH	N1	S1	F2	Portfolio
portfolio-details.html	FARAH	N1	S1	F2	Portfolio
pulmotrack-details.html	FARAH	N1	S1	F2	Portfolio
testpilot-details.html	FARAH	N1	S1	F2	Portfolio
digital-gallery.html	FARAH	N1	S1	F1	Digital Gallery
contact.html	FARAH	N1	S1	F2	Contact
starter-page.html	FARAH	N1	S1	F3	(none)

Navigation N1

Exact label (in source order)	Destination
Home	index.html
About	about.html
Resume	resume.html
Certifications	certifications.html
Services	services.html
Portfolio	portfolio.html
Digital Gallery	digital-gallery.html
Contact	contact.html

Header social links S1

Label / icon class	Exact destination
GitHub	https://github.com/FarahSayari
Email	mailto:farahsayari.dev@gmail.com
LinkedIn	https://www.linkedin.com/in/farah-sayari-060745248

Footer F1

> © Copyright Farah Sayari All Rights Reserved

Social label / icon class	Exact destination
GitHub	https://github.com/FarahSayari
Email	mailto:farah.sayari@esprit.tn
LinkedIn	https://www.linkedin.com/in/farah-sayari-060745248

Footer F2

> © Copyright Farah Sayari All Rights Reserved

Social label / icon class	Exact destination
GitHub	https://github.com/FarahSayari
Email	mailto:farahsayari.dev@gmail.com
LinkedIn	https://www.linkedin.com/in/farah-sayari-060745248

Footer F3

> © Copyright Kelly All Rights Reserved

Social label / icon class	Exact destination
Icon: bi bi-twitter-x	(empty)
Icon: bi bi-facebook	(empty)
Icon: bi bi-instagram	(empty)
Icon: bi bi-linkedin	(empty)

An empty destination is shown as “(empty)”; a destination of # is a fragment placeholder. The starter template contains a malformed ThemeWagon anchor in its source. Its browser-parsed label and destination above are recorded as they currently resolve.

Shared interactions and appearance

The site has a sticky header, a mobile navigation toggle, an initial preloader, a scroll-to-top icon, and animation-on-scroll effects. These elements have no additional visible body copy. The scroll-to-top icon uses href="#" and is activated after scrolling past 100 pixels in the current shared script.

Style token	Current value
default-font	"Roboto", system-ui, -apple-system, "Segoe UI", Roboto, "Helvetica Neue", Arial, "Noto Sans", "Liberation Sans", sans-serif, "Apple Color Emoji", "Segoe UI Emoji", "Segoe UI Symbol", "Noto Color Emoji"
heading-font	"Raleway", sans-serif
nav-font	"Poppins", sans-serif
background-color	#ffffff
default-color	#444444
heading-color	#222222
accent-color	#34b7a7

01 · Home

Page attribute	Exact value
Source file	index.html
Browser title	Farah Sayari Portfolio
Language	en
Description meta tag	(empty)
Keywords meta tag	(empty)
Main section IDs	hero

Exact page content

Farah Sayari

> I'm a Software Engineer|

> Welcome to my digital world!

> About Me

> View Resume

Animated content and alternate states

The fixed introduction is “I'm a”. The animated span cycles through these exact titles; the cursor displays “|”. The first complete title is used in the transcript above.

> Software Engineer

> Data Scientist

> Digital Artist

Animation parameter	Current value
typingSpeed	110 ms
deletingSpeed	70 ms
pauseAfterTyping	1400 ms
pauseAfterDeleting	300 ms

Main-content links and downloads

Exact label / accessible label	Exact destination	Download filename
About Me	about.html	(empty)
View Resume	resume.html	(empty)

Main-content images

Image path	Exact alternate text	Role
assets/img/hero-bg.jpg	Abstract hero background for Farah Sayari	(empty)

02 · About

Page attribute	Exact value
Source file	about.html
Browser title	About - Farah Sayari
Language	en
Description meta tag	(empty)
Keywords meta tag	(empty)
Main section IDs	about

Exact page content

About

> Get to know me, my background, and the kind of digital solutions I enjoy building.

Junior Software Engineer specialized in Data Science

> Hi, I’m Farah Sayari. I’m a Software Engineer specialized in Data Science, a book lover, and a creative soul with a passion for digital art.

> I build smart, practical, and user-friendly solutions using AI, Machine Learning, Deep Learning, automation, and full-stack development. Logic helps me build. Imagination helps me create.

- Email: farahsayari.dev@gmail.com
- Phone: +216 28528390
- Linkedin: linkedin.com/in/farah-sayari-060745248
- Degree: Software Engineering – Data Science
- Github: github.com/FarahSayari
- Freelance: Open to opportunities

> I’m open to junior opportunities in Software Engineering, Data Science, and AI.

Animated content and alternate states

The 3 profile images alternate in a continuous loop. Current switch interval: 500 ms. Image paths and exact alternate text are in the image inventory below.

Main-content links and downloads

Exact label / accessible label	Exact destination	Download filename
farahsayari.dev@gmail.com	mailto:farahsayari.dev@gmail.com	(empty)
linkedin.com/in/farah-sayari-060745248	https://www.linkedin.com/in/farah-sayari-060745248	(empty)
github.com/FarahSayari	https://github.com/FarahSayari	(empty)

Main-content images

Image path	Exact alternate text	Role
assets/img/me11.png	Farah Sayari profile photo 1	(empty)
assets/img/me22.png	Farah Sayari profile photo 2	(empty)
assets/img/me33.png	Farah Sayari profile photo 3	(empty)

03 · Resume

Page attribute	Exact value
Source file	resume.html
Browser title	Resume - Farah Sayari
Language	en
Description meta tag	Explore Farah Sayari's resume: AI and software engineering experience, education, technical skills, and projects in Data Science and agentic AI.
Keywords meta tag	(absent)
Main section IDs	resume

The source has two columns: summary, education, skills, languages, and activities first; professional experience and projects second. The text below follows that source order. The downloadable CV is a separate PDF and is appended to the final PDF reference unchanged.

Exact page content

Resume

> AI & Software Engineer specializing in Data Science, agentic AI, and full-stack development.

> Download CV (PDF)

Summary

Farah Sayari

> Software Engineer specializing in Data Science, with hands-on experience in Machine Learning, Deep Learning, agentic AI, and full-stack development. I build intelligent, data-driven applications and am seeking a full-time opportunity in AI, Data Science, or Software Engineering.

- Ariana, Tunisia
- +216 28 528 390
- farahsayari.dev@gmail.com
- GitHub · LinkedIn

Education

Engineering Cycle in Computer Science – Data Science

> Sep 2023 – Nov 2026

> ESPRIT – Private Higher School of Engineering and Technology

Bachelor’s Degree in Computer Systems Engineering – Embedded Systems & IoT

> Sep 2020 – Jun 2023

> Faculty of Sciences of Bizerte

Technical Skills

Programming

> Python, Java, JavaScript, R, C

AI & Machine Learning

> Scikit-learn, TensorFlow, Keras, PyTorch, MLflow, Deep Learning, NLP, Recommendation Systems

Generative AI & Agents

> LLMs, RAG, LangChain, LangGraph, Qdrant, Ollama, Prompt Engineering

Backend & Frontend

> Spring Boot, FastAPI, Flask, Angular, React, REST APIs

Data Engineering & Analytics

> SQL, PostgreSQL, Hadoop, Talend, Data Warehousing, Power BI, Time Series Analysis

DevOps & Tools

> Git, Docker, GitHub Actions, Jenkins, Linux, Cypress, Kubernetes

Languages

- Arabic Native
- French C2
- English C1

Activities & Awards

- IEEE Computational Intelligence Society (CIS) — ESPRIT Chapter, 2025.
- Bal des Projets — ESPRIT (2024 & 2025) — Academic projects pre-selected for both editions.
- IEEEXtreme 17.0 & 18.0 — Participant.
- University English Competition (2024) — 3rd Place.

> Interests: Reading & Literature, Digital Art & Illustration.

Professional Experience

AI & Full-Stack Software Engineer Intern

> Feb – Aug 2026

> VERMEG · PFE Internship

> TestPilot — AI-Assisted Test Automation Platform

- Designed and developed a full-stack test automation platform using Angular, Spring Boot, FastAPI, and PostgreSQL, enabling QA teams to manage, generate, and execute automated tests.
- Engineered a hybrid agentic AI pipeline using Python, LangGraph, and LLMs to transform recorded user interactions into structured test models and executable Cypress scripts.
- Developed a Chrome extension (Manifest V3) to capture browser interactions and implemented LLM-assisted script repair with validation and controlled retries.
- Integrated Cypress test execution, automated reporting, GitHub Actions CI/CD, and Docker to support continuous testing and deployment.

Data Scientist Intern

> Jul – Aug 2025

> Esprit-Tech · Summer Internship

> Clustering Algorithms for Heterogeneous Student Groups

- Implemented and compared multiple clustering algorithms, achieving a 0.72 Silhouette Score, and built an interactive Streamlit dashboard to visualize and evaluate results.
- Developed an automated group formation pipeline integrated into the backend, supporting teachers in forming balanced and diverse student teams.

Projects

LollyBooks

> Oct – Present · Ongoing

> Co-Founder & Software Engineer

> Personalized Digital Library & AI Reading Companion

- Co-designed and developed an interactive digital library featuring customizable bookshelves, reading analytics, and goal tracking.
- Integrated the Open Library REST API and Goodreads CSV processing to automate book metadata retrieval and reading history imports.
- Prototyped recommendations based on reading preferences and mood, and designed an LLM-powered reading companion for conversational book discovery.

Aura

> Oct – Present · Ongoing

> AI & Software Engineer

> Agentic AI Career Assistant

- Redesigning a local, agentic AI assistant to automate job discovery and match job descriptions with personal skills and experience.
- Developing an LLM-powered application assistant to generate tailored cover letters, analyze CV compatibility, and recommend targeted resume improvements.
- Adapting RAG, semantic search, and agent orchestration using LangChain, LangGraph, and Qdrant to support personalized job search and application workflows.

Main-content links and downloads

Exact label / accessible label	Exact destination	Download filename
Download CV (PDF)	assets/docs/Farah_Sayari_CV2026.pdf	Farah_Sayari_CV2026.pdf
+216 28 528 390	tel:+21628528390	(empty)
farahsayari.dev@gmail.com	mailto:farahsayari.dev@gmail.com	(empty)
GitHub	https://github.com/FarahSayari	(empty)
LinkedIn	https://www.linkedin.com/in/farah-sayari-060745248	(empty)

04 · Certifications

Page attribute	Exact value
Source file	certifications.html
Browser title	Certifications - Farah Sayari
Language	en
Description meta tag	Courses and certifications from my learning journey.
Keywords meta tag	(empty)
Main section IDs	certifications

Exact page content

Certifications

> Courses and certifications from my learning journey.

Always learning.

> A place to share the certificates I earn through courses and continued learning.

> My course certificates are coming soon.

05 · Services

Page attribute	Exact value
Source file	services.html
Browser title	Services - Farah Sayari
Language	en
Description meta tag	(empty)
Keywords meta tag	(empty)
Main section IDs	services

Exact page content

Services

> Here are the areas where I enjoy building, experimenting, and creating digital solutions.

Data Science & AI

> Building intelligent solutions with machine learning, deep learning, data analysis, and AI-powered workflows.

Web Development

> Creating practical and user-friendly web applications using modern frontend and backend technologies.

AI Agents & Automation

> Designing smart agents, RAG pipelines, and automated workflows that help reduce manual tasks and improve decision-making.

Test Automation & QA

> Working on automated testing workflows, test generation, execution, and reporting to improve software reliability.

Data Analytics & BI

> Transforming raw data into clear insights using dashboards, reporting tools, statistics, and business intelligence techniques.

Creative Digital Work

> Exploring digital art, visual design, and friendly UI ideas to make technology feel more human and engaging.

> I like combining logic, data, creativity, and usability to build solutions that are both smart and pleasant to use.

06 · Portfolio

Page attribute	Exact value
Source file	portfolio.html
Browser title	Portfolio - Farah Sayari
Language	en
Description meta tag	Explore Farah Sayari's projects in software engineering, Data Science, and AI, starting with LollyBooks.
Keywords meta tag	(absent)
Main section IDs	portfolio

The portfolio uses a masonry grid with project labels in hover overlays. The image inventory below records the current thumbnails. Entries with no description or detail links are recorded as placeholders.

Exact page content

Portfolio

> A selection of my work in software engineering, Data Science, and AI.

- All
- App
- Web

LollyBooks

> Personalized Digital Library & AI Reading Companion

AI Agent Recruiter

TestPilot

> AI-Assisted Test Automation Platform

PulmoTrack

> AI-Powered Pulmonary Nodule Analysis

Project entries and filter membership

Project	Description	Filters	Detail destination
LollyBooks	Personalized Digital Library & AI Reading Companion	App, Web	portfolio-details.html

Project	Description	Filters	Detail destination
AI Agent Recruiter	(empty)	App	(none)
TestPilot	AI-Assisted Test Automation Platform	App	testpilot-details.html
PulmoTrack	AI-Powered Pulmonary Nodule Analysis	Web	pulmotrack-details.html

The default filter is All. Filter membership and every existing detail destination are listed in the table above. Project image preview links open the lightbox. Entries without detail links remain unlinked. The filter controls also support Enter and Space in the current inline script.

Main-content links and downloads

Exact label / accessible label	Exact destination	Download filename
Preview LollyBooks artwork	assets/img/portfolio/lollybooks/library-art.jpg	(empty)
LollyBooks project details	portfolio-details.html	(empty)
TestPilot project details	testpilot-details.html	(empty)
PulmoTrack project details	pulmotrack-details.html	(empty)

Main-content images

Image path	Exact alternate text	Role
assets/img/portfolio/lollybooks/library-art.jpg	Cozy pink bookshelves with string lights and a My Reading Library neon sign	(empty)
assets/img/masonry-portfolio/masonry-portfolio-2.jpg	(empty)	presentation
assets/img/portfolio/testpilot/TESTPILOT%20AI%20Automation%20Platform.png	TestPilot logo and AI-Assisted Test Automation Platform title on a purple gradient background	(empty)
assets/img/portfolio/pulmotrack/pulmotrack.png	PulmoTrack project artwork illustrating CT-based nodule detection, segmentation, and classification	(empty)

07 · LollyBooks project details

Page attribute	Exact value
Source file	portfolio-details.html
Browser title	LollyBooks - Farah Sayari
Language	en
Description meta tag	LollyBooks: a personalized digital library and AI reading companion. Discover Farah Sayari's contributions as Co-Founder and Software Engineer.
Keywords meta tag	(absent)
Main section IDs	portfolio-details

Exact page content

> Back to Portfolio

LollyBooks — Personalized Digital Library & AI Reading Companion

> A cozy digital space to track your reading journey, discover new books, and build a library that reflects your personality.

Project Information

- Category: Web Application · Personalization & Recommendation Systems
- Project Type: Personal Startup Project
- Role: Co-Founder & Software Engineer
- Project Date: October 2026 – Present
- Status: Active Development (Functional Prototype)
- Project URL: Coming Soon

Your Library, Your Stories, Your Little World

> LollyBooks is a personalized digital library designed to make reading tracking more engaging, accessible, and enjoyable.

> The idea was born from a personal passion for reading and the desire to create a beautiful virtual bookshelf where readers can preserve their reading journey, even without owning physical copies of their books.

> As a Co-Founder and Software Engineer, I contributed to the product vision, interface design, and development of the initial web application prototype, combining interactive book management, reading analytics, and personalized book discovery.

Key Features & Contributions

- Interactive Digital Library: Developed customizable virtual bookshelves with multiple display modes, allowing readers to organize their collections visually.
- Book Data Integration: Integrated the Open Library REST API for automated metadata retrieval and implemented Goodreads CSV importing to migrate reading histories, ratings, and reviews.
- Reading Analytics & Goals: Implemented reading statistics, progress tracking, and personalized reading goals to help users reflect on their reading habits.
- Personalized Recommendations: Prototyped a recommendation system based on reading history, ratings, genre preferences, and reading moods.
- Reading Companion: Designed a conversational recommendation experience to help readers discover books aligned with their interests and preferences.

Project Vision & Future Development

> LollyBooks is evolving from a functional prototype into a scalable, AI-powered reading platform.

> The long-term vision includes an LLM-powered reading companion, deeper personalization through reading reviews and preferences, multilingual support, community features, and a production-ready architecture.

> The goal is to create more than a book-tracking application: a personal reading universe that grows and evolves with every reader.

Main-content links and downloads

Exact label / accessible label	Exact destination	Download filename
Back to Portfolio	portfolio.html	(empty)

Main-content images

Image path	Exact alternate text	Role
assets/img/portfolio/lollybooks/library-art.webp	Cozy pink bookshelves with string lights and a My Reading Library neon sign	(empty)

08 · PulmoTrack project details

Page attribute	Exact value
Source file	pulmotrack-details.html
Browser title	PulmoTrack - Farah Sayari
Language	en
Description meta tag	Explore Farah Sayari's PulmoTrack LUNA16 CT research: candidate screening, true/false nodule classification, exploratory segmentation, notebook evaluations, and a live Streamlit showcase.
Keywords meta tag	(absent)
Main section IDs	portfolio-details, research-paper

Exact page content

> Back to Portfolio

PulmoTrack — Deep Learning for Pulmonary Nodule Analysis

> Research · Deep Learning · Medical Imaging · Computer Vision

> An academic medical-imaging project using LUNA16 CT scans to study nodule candidate detection, binary classification and exploratory 3D segmentation, presented through an interactive research website.

> Previous image

> Next image

Project Information

- Category: Research, DL, Medical Imaging, Computer Vision
- Role: Data Scientist, Project Manager
- Methodology: CRISP-DM
- Dataset: LUNA16
- Status: Completed
- Scientific Paper: Available · Team research manuscript

> Open Live Showcase (opens in a new tab)

> View Source on GitHub (opens in a new tab)

Technologies

> Research: Python, PyTorch, torchvision, NumPy, pandas, SciPy, scikit-learn, SimpleITK.

> Analysis & delivery: Jupyter Notebook, Google Colab, Matplotlib, Plotly, Pillow, Streamlit, Git, GitHub and Streamlit Community Cloud.

Project Summary

> Jump directly to a section of the project.

- Project Information
- Technologies
- Project Overview
- My Contributions
- LUNA16 Dataset
- Project Methodology
- Explore the Three DSOs DSO1 — Nodule Detection Data Processing Models Applied Results & Evaluation DSO2 — Nodule Classification Data Processing Models Applied Results & Evaluation DSO3 — Exploratory 3D Segmentation Data Processing Models Applied Results & Evaluation
- Deployment
- Research Paper Paper-reported LUNA16 highlights

Project Overview

> Lung cancer is the leading cause of cancer-related death worldwide and is often diagnosed at an advanced stage, when treatment options are more limited. Earlier identification and appropriate follow-up can improve the opportunity for effective treatment.

> PulmoTrack explores how deep learning can analyze pulmonary nodules in volumetric CT data. The project connects medical-image preprocessing, three research objectives, model evaluation and a deployed interface that makes the experiments understandable to a portfolio audience.

> The work is an academic research study. Its recorded models evaluate nodule candidates and masks; they do not establish a clinically validated cancer diagnosis system.

> Scientific context: World Health Organization — Lung cancer.

My Contributions

- LUNA16 preprocessing. Worked on CT resampling, annotation and candidate inspection, world-to-voxel coordinate conversion, intensity normalization and 3D patch preparation.
- Detection and classification. Developed and studied 3D candidate-screening and binary classification approaches, comparing their saved training histories, confusion matrices and ROC evaluations.
- Exploratory segmentation. Investigated 3D U-Net, attention and residual-attention variants, and tiny-nodule oversampling, documenting both promising summaries and difficult positive-only cases.
- Results and documentation. Organized notebook outputs, kept evaluation cohorts and checkpoint results distinct, and contributed to the team research manuscript.

- Deployment. Created a Streamlit research showcase with CT visualizations, model overviews, result interpretation, source resources and access to the research paper.

LUNA16 Dataset

> LUNA16 (Lung Nodule Analysis 2016) is derived from LIDC-IDRI and contains 888 volumetric chest CT scans. It provides CT volumes, candidate locations and nodule annotations for pulmonary nodule research.

> The official reference standard includes nodules at least 3 mm in diameter accepted by at least three of four radiologists. Scans with slice thickness greater than 2.5 mm were excluded. The CT collection is approximately 90 GB in scale; the deployed showcase uses saved outputs rather than distributing the dataset.

> The project prepares different patch collections for different experiments: 64^3 voxels for detection/classification and 48^3 voxels for the archived segmentation studies.

> Dataset source and access: LUNA16 official data page.

Project Methodology

> PulmoTrack follows the CRISP-DM methodology, from understanding the research problem and preparing LUNA16 CT data to modeling, evaluation, and deployment. This iterative workflow guides the three research objectives and the presentation of their results in the Streamlit showcase.

Explore the Three DSOs

> Data processing prepares the CT scans for learning; modeling builds and trains networks using that prepared data. Before preparing the task-specific datasets, we resampled CT volumes to 1 × 1 × 1 mm voxel spacing and converted annotation coordinates from millimetres to voxel locations using each scan's origin and spacing, accounting for medical-image order (x, y, z) and NumPy array order (z, y, x).

> The evaluation tables preserve each saved checkpoint's recorded cohort. Accuracy is calculated from the original confusion matrix, while ROC-AUC comes from the corresponding recorded output.

DSO1 — Nodule Detection

> This objective screens supplied candidate locations. Each input is a 3D patch around a candidate, and the output is a nodule or non-nodule label.

Data Processing

- Candidate matching: Compared candidates with annotated nodule centres from the same scan. A distance below 5 mm defined the strict positive label for the selected dataset; preparation also explored a looser 10 mm label.
- Dataset selection: Retained 1,215 positive candidates and randomly sampled 8,000 negatives, producing 9,215 patches. The larger negative collection exposes the model to varied non-nodule appearances.
- Patch extraction: Extracted 64 × 64 × 64 patches around candidates and padded border patches to preserve the expected dimensions.
- Normalization and tensors: Subtracted each patch's mean and divided by its standard deviation, then created one-channel tensors with shape (1, 64, 64, 64).
- Dataset split: Used a seeded random patch split of 6,451 training, 1,843 validation and 921 test examples.

Models Applied

> Applied a U-Net-style 3D encoder classifier with 32 → 64 → 128 → 256 feature channels. Convolution and pooling build 3D features, followed by global average pooling and a linear head that produces two class scores. This model has no segmentation decoder and predicts a patch label, rather than a boundary or bounding box.

> Training used weighted focal loss (γ = 2), Adam with learning rate 0.0001, batch size 16 and 20 recorded epochs. Class weights emphasize less frequent positive examples, while focal loss concentrates learning on difficult cases. The lowest validation loss determined the selected checkpoint.

Results & Evaluation

Cohort	Patches	Accuracy	ROC-AUC
Validation	1,843	98.37%	0.9928
Test	921	97.83%	0.9961

> DSO1 — Test-set confusion matrix.

> DSO1 — Test-set ROC curve; AUC is rounded in the plot legend.

> DSO1 — Training and validation history.

> On the 921-patch test cohort, the saved model correctly identifies 112 of 121 nodule patches (92.56% sensitivity), with 11 false positives and 9 false negatives. These are candidate-level outcomes; they do not establish false positives per full CT scan.

> Interpretation: The test accuracy of 97.83% and ROC-AUC of 0.9961 show strong separation on the recorded candidate cohort. Sensitivity of 92.56% also makes the remaining nine missed nodule patches visible.

DSO2 — Nodule Classification

> The recorded task classifies a 3D candidate patch as a true nodule or false nodule. Although the wider project discusses benign/malignant classification, these implemented evaluations use binary candidate labels.

Data Processing

- Separate patch collection: Prepared this dataset independently of the 9,215 detection examples. The preparation cell filters candidates marked as class 1 and compares their positions with annotation centres using a 6 mm matching threshold.
- Patch extraction and normalization: Extracted 64 × 64 × 64 patches, padded when needed, and applied mean-and-standard-deviation normalization.
- Labels and storage: Saved tensors into separate folders, with label 1 for true nodules and 0 for false nodules.
- Dataset and initial split: The notebook reports 1,243 true-nodule and 761 false-nodule patches, totalling 2,004. The initial split contains 1,603 training, 200 validation and 201 test patches.

Models Applied

- Improved 3D CNN: Three convolution blocks followed by adaptive pooling, a dense layer, dropout and a two-class output; 17 recorded epochs.

- ResNet3D Lite: Residual blocks combine transformed features with shortcut connections, followed by global pooling and a classification head; 6 recorded epochs.
- U-Net encoder classifier: Four encoder stages followed by global pooling and a dense two-class head; 30 recorded epochs.

> All three architectures use cross-entropy loss and Adam with an initial learning rate of 0.0001. Dropout regularizes the CNN and U-Net heads. Validation loss guides checkpoint selection, and saved evaluations are kept separate from final-epoch logs.

Results & Evaluation

Study / cohort	Patches	Accuracy	ROC-AUC
Improved 3D CNN · validation	200	74.50%	0.8269
Improved 3D CNN · test	201	74.63%	0.8335
ResNet3D Lite · validation	401	72.07%	0.7678
U-Net encoder · validation	200	74.00%	0.8116

> DSO2 — Improved 3D CNN training and validation history.

> The Improved 3D CNN test evaluation identifies 107 of 123 true-nodule patches, with 35 false positives and 16 false negatives. Its accuracy and AUC describe binary candidate labels, not malignancy diagnosis.

> How to compare these results. Each model retains its own recorded cohort. The ResNet matrix contains 401 patches despite an earlier 200-patch validation log. Splitting is by patch, and a patient-disjoint benchmark is not established. These studies support analysis of the recorded runs, not a controlled architecture ranking.

> Interpretation: The Improved 3D CNN reaches 74.63% test accuracy and ROC-AUC 0.8335 on its 201-patch cohort. Its 35 false positives and 16 false negatives show the remaining classification errors. The differing evaluation cohorts prevent treating these recorded runs as a controlled ranking of the three architectures.

DSO3 — Exploratory 3D Segmentation

> This objective predicts a mask identifying nodule voxels within a 3D CT patch. It is a parallel research exploration, rather than a prerequisite for detection or classification.

Data Processing

- Volume preparation: Prepared lung regions, clipped intensities to −1000 to 400 HU and scaled them to the range [0, 1].
- Reference masks: Created spherical masks from annotated nodule centres and diameters. These approximate reference regions are not manually traced nodule boundaries.
- Patch extraction: Extracted 48 × 48 × 48 positive CT patches with corresponding mask patches. Negative patches were sampled away from annotated nodules and paired with empty masks.
- Augmentation and normalization: Explored flips, rotations, zoom, noise and intensity shifts, with patch-level min–max normalization in the model loader.

- Splits and tiny nodules: The baseline records 2,036 training and 509 validation patches. Later experiments oversampled selected small-nodule examples to increase their exposure during training.

Models Applied

> The segmentation flow is CT patch → encoder → bottleneck → decoder → voxel probability map. Decoder upsampling and encoder-to-decoder skip connections help reconstruct spatial detail.

- 3D U-Net: Baseline encoder–decoder architecture.
- Attention 3D U-Net: Attention gates weight features passed through skip connections.
- Residual Attention 3D U-Net: Residual convolution blocks combined with attention gates.
- Tiny-nodule training: Baseline architecture with modified sampling and augmentation.

> Recorded training used binary cross-entropy loss and Adam at 0.0001. A sigmoid produces voxel probabilities, and a 0.5 threshold converts them into binary masks.

Results & Evaluation

> Segmentation results depend strongly on the selected patches and the handling of empty masks. The table preserves these separate experimental contexts; all values are notebook-reported summaries.

Model study	Evaluation context	Patches	Reported score
3D U-Net	Five positives · A	5	Dice 0.2115
3D U-Net	Five positives · B	5	Dice 0.3127
3D U-Net	Mixed validation loader	509	Dice 0.9504
Attention U-Net	Five positive patches	5	Dice 0.1599
Attention U-Net	Ten mixed patches	10	Dice 0.9800
Residual attention	Ten positives · initial	10	Dice 0.1200
Residual attention	Ten positives · later	10	Dice 0.0348

> DSO3 — Baseline 3D U-Net training and validation loss.

> The 0.9504 validation Dice comes from a mixed, augmented loader, where smoothing assigns empty prediction/reference pairs a Dice near 1. The attention model's mixed-patch evaluator similarly assigns perfect scores to empty pairs. Positive-only samples show the remaining difficulty of recovering small nodule regions.

> Tiny-nodule augmentation retains saved prediction examples and a final validation binary cross-entropy loss of 0.0446. No Dice, precision, recall or F1 evaluation was recorded for that run; a lower loss is not an overlap score.

> Recorded Attention 3D U-Net positive-nodule example — Dice 0.8000, precision 0.8630, recall 0.7456. Single case from the ten-patch mixed evaluation.

> Evaluation context: The notebook applies random augmentation to images and masks through separate calls, so their spatial transformations are not guaranteed to match; validation also inherits augmentation. Patch splits do not establish patient-disjoint evaluation. Together with the mixed-versus-positive-only score differences,

> these limitations keep the segmentation findings exploratory.

Deployment

> My PulmoTrack contribution is hosted on Streamlit Community Cloud, with the GitHub repository as its deployment source. The interface preserves the dark navy and teal medical-imaging identity, sidebar navigation and CT-focused visual presentation.

> Visitors can explore model architectures, recorded metrics, training histories, original segmentation predictions, pilot Grad-CAM, research resources and the manuscript viewer. The app presents saved evidence; it requires no CT dataset download or live model execution.

> Delivery workflow: Research notebooks → exported figures and result summaries → Streamlit interface → GitHub → Streamlit Community Cloud.

> The website is a completed research showcase, rather than a patient-upload or diagnostic inference service. Repository changes can be reflected in the deployed app through Community Cloud's GitHub integration.

> Open Live Showcase (opens in a new tab)

> View Source on GitHub (opens in a new tab)

> Deployment reference: Streamlit Community Cloud documentation.

Research Paper

> Title: PulmoTrack: A Deep Learning Solution for Pulmonary Nodule Segmentation, Detection, and Classification on CT and Chest X-ray Scans

> Document: Team research manuscript · May 2025 · 11 pages.

> Authors: Farah Sayari and co-authors.

> The manuscript covers the broader team project, including CT and chest X-ray work. This portfolio page focuses on the LUNA16 CT contribution.

Paper-reported LUNA16 highlights

> The manuscript reports segmentation Dice 0.95, detection sensitivity 92% at 1.5 false positives per scan, and classification AUC 0.91 (pp. 4–5).

> View Paper (PDF) (opens in a new tab)

> Download Paper (PDF)

Animated content and alternate states

The project image card contains 2 manually switchable images. Previous/next buttons, slide indicators, keyboard navigation, and touch swipes select an image. Automatic rotation is disabled; the first image is initially selected.

Slide	Image path
1	assets/img/portfolio/pulmotrack/pulmotrack.png
2	assets/img/portfolio/pulmotrack/team.png

Main-content links and downloads

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Enlarge the PulmoTrack team photo	assets/img/portfolio/pulmotrack/team.png	(empty)
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View Source on GitHub (opens in a new tab)	https://github.com/FarahSayari/pulmoTrack-farahsayari-contribution	(empty)
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Project Overview	#project-overview	(empty)
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Results & Evaluation	#detection-results-title	(empty)
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Results & Evaluation	#classification-results-title	(empty)
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Paper-reported LUNA16 highlights	#paper-reported-highlights	(empty)
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LUNA16 official data page	https://luna16.grand-challenge.org/Data/	(empty)
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Streamlit Community Cloud documentation	https://docs.streamlit.io/deploy/streamlit-community-cloud/manage-your-app	(empty)
View Paper (PDF) (opens in a new tab)	assets/docs/PulmoTrackPaper.pdf	(empty)
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Main-content images

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assets/img/portfolio/pulmotrack/Nodule%20classification.png	PulmoTrack classification concept artwork depicting the broader benign/malignant project goal; the recorded evaluations below use true/false nodule labels	(empty)

Image path	Exact alternate text	Role
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assets/img/portfolio/pulmotrack/08_dso2_training_history.png	Improved 3D CNN training and validation loss and accuracy across 17 epochs	(empty)
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assets/img/portfolio/pulmotrack/arch%20segmentation.png	3D U-Net segmentation architecture: CT preprocessing, patch extraction, encoder, bottleneck, decoder with skip connections, and output nodule masks	(empty)
assets/img/portfolio/pulmotrack/09_dso3_training_loss.png	3D U-Net training and validation loss across 50 epochs	(empty)
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09 · TestPilot project details

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Keywords meta tag	(absent)
Main section IDs	portfolio-details

Exact page content

> Back to Portfolio

TestPilot — AI-Assisted Test Automation Platform

> Full-Stack Engineering · Agentic AI · Test Automation · DevOps

> Developed during my end-of-studies internship (PFE) at VERMEG, TestPilot is an AI-assisted regression testing platform that transforms recorded browser interactions into structured test scenarios and automated Cypress tests, integrating AI-powered generation, human validation, and test management.

> Previous image

> Next image

Project Information

- Category: Software Engineering · Agentic AI · Test Automation
- Project Type: End-of-Studies Internship (PFE)
- Company: VERMEG
- Role: Full-Stack & AI Software Engineer
- Duration: February – August 2026
- Methodology: Scrumban
- Status: Completed — Functional Prototype

Technologies & Tools

> Frontend: Angular

> Backend: Java · Spring Boot · Spring Security · REST APIs

> AI & Agentic Workflows: Python · FastAPI · LangGraph · LLMs · Ollama · Gemma

> Browser Automation: Chrome Extension (Manifest V3) · Cypress · JavaScript

> Database: PostgreSQL

> DevOps & Infrastructure: Docker · Nginx · GitHub Actions · CI/CD

Project Summary

> Jump directly to a section of the project.

- Project Information
- Technologies & Tools
- Project Overview
- The Challenge
- The Solution End-to-End Workflow
- Development Methodology — Scrumban
- My Contributions Full-Stack Application Development Browser Recording & Test Automation Agentic AI Pipeline DevOps & Integration
- AI Architecture — Controlled Agentic Automation Agent 1 — Scenario Processing Agent 2 — Test Model Generation Agent 3 — Cypress Script Generation
- Additional Capabilities Test Management AI Assistance Chatbot CI/CD Integration
- Demonstrations & Validation Demo 1 — From Scenario Recording to Automated Testing Demo 2 — CI/CD-Triggered Regression Testing Demonstration Application — SauceDemo
- Key Takeaways

Project Overview

> Developed during my end-of-studies internship at VERMEG, TestPilot is a full-stack, AI-assisted platform designed to streamline regression testing.

> It enables Business Analysts and QA Engineers to record application workflows, transform them into structured test models, and generate automated Cypress scripts through a hybrid agentic AI pipeline.

> The platform goes beyond test generation by integrating test management, script validation, human approval, automated execution, and reporting.

> Through Docker and GitHub Actions, TestPilot also supports continuous testing within CI/CD workflows, connecting test automation to the software delivery process.

> The objective is to make regression testing more efficient, reliable, and traceable while keeping human oversight at the center of the process.

The Challenge

> In complex software environments, frequent updates make regression testing essential to ensure applications remain stable and reliable.

> Although test automation tools already exist, creating and maintaining test scripts still requires significant manual effort. QA teams must write scripts, update them as applications evolve, and manage test executions.

> This repetitive work takes time, increases the risk of human errors, and can slow down software delivery. AI can help automate these tasks, but generated tests still need to be reviewed and validated.

> The challenge: How can we reduce the manual effort of creating and maintaining automated regression tests while ensuring their reliability and keeping human control?

The Solution

> TestPilot connects the different stages of regression testing in a single platform, combining browser recording, AI-assisted generation, controlled validation, and automated execution.

> Rather than relying entirely on AI-generated code, the solution follows a structured process in which test assets are progressively generated, validated, and reviewed.

End-to-End Workflow

- Record: Capture real user interactions through a custom Chrome extension.
- Process: Transform recorded actions into structured scenarios, variables, and reliable locators.
- Generate: Produce a Test Model and an executable Cypress script draft.
- Validate: Verify the generated script through automated execution.
- Review & Approve: Allow QA teams to inspect and approve validated scripts.
- Manage & Execute: Organize regression campaigns through Test Plans and Test Cycles, then analyze execution results.

> This workflow helps bridge the gap between business requirements and executable automated tests while maintaining human control over the final testing process.

Development Methodology — Scrumban

> TestPilot was developed using Scrumban, combining the iterative approach of Scrum with the flexibility of Kanban.

> Development was organized around a prioritized backlog, incremental feature delivery, and regular progress reviews.

> The project evolved through successive stages, from the full-stack foundation and browser recording extension to the AI generation pipeline, test management, validation, and CI/CD integration.

> This approach allowed me to continuously test new features, address technical challenges, and refine the platform based on feedback throughout the internship.

My Contributions

> I was responsible for designing and developing the platform across its principal components.

Full-Stack Application Development

- Developed the web application using Angular, Spring Boot, and PostgreSQL.
- Implemented REST APIs, authentication, role-based access control, and project management features.
- Built interfaces for scenario recording, test management, script validation, and execution reporting.

Browser Recording & Test Automation

- Developed a Chrome extension using Manifest V3 to capture browser interactions and relevant page observations.
- Implemented action processing, locator selection, and variable extraction to reconstruct recorded workflows.
- Integrated Cypress for automated test validation and regression execution.

Agentic AI Pipeline

- Designed a hybrid architecture combining deterministic processing with controlled LLM assistance.
- Developed workflows for scenario processing, Test Model generation, and Cypress script generation.
- Integrated AI-assisted script improvement with validation feedback and bounded execution controls.

DevOps & Integration

- Containerized application services using Docker.
- Integrated GitHub Actions for CI/CD workflows.
- Connected automated regression execution with reporting and traceability mechanisms.

AI Architecture — Controlled Agentic Automation

> At the core of TestPilot is a hybrid agentic AI pipeline designed to transform recorded browser interactions into automated tests.

> The architecture separates the generation process into three specialized stages:

Agent 1 — Scenario Processing

> Cleans and structures recorded interactions, extracts variables, and identifies suitable element locators.

Agent 2 — Test Model Generation

> Transforms processed actions into a structured Test Model containing test steps, preconditions, variables, and proposed assertions.

Agent 3 — Cypress Script Generation

> Converts the Test Model into an executable Cypress test script draft.

> The pipeline combines deterministic processing with controlled LLM assistance, allowing AI to support generation and improvement without independently approving tests.

> Generated scripts undergo automated validation and human review before being approved for official test execution.

Additional Capabilities

Test Management

> TestPilot includes a test management module for organizing Test Cases, Test Plans, and Test Cycles, allowing QA teams to manage regression campaigns and follow execution results.

AI Assistance Chatbot

> An integrated assistant supports users by answering questions about testing workflows, application features, and authorized project information using documentation retrieval and controlled read-only access.

CI/CD Integration

> GitHub Actions integration demonstrates how regression campaigns can be triggered within a software delivery workflow, with execution results retrieved for reporting.

Demonstrations & Validation

> To demonstrate TestPilot's capabilities, I used SauceDemo, an external sample e-commerce application, as a controlled testing environment.

> The demonstrations cover the complete testing workflow, from recording user interactions and generating automated tests to triggering regression campaigns through CI/CD.

Demo 1 — From Scenario Recording to Automated Testing

> This demonstration illustrates how TestPilot transforms a recorded business scenario into an executable automated test.

> It covers:

- Recording browser interactions using the Chrome extension.
- Processing the recorded actions.
- Generating the structured Test Model.
- Generating the Cypress script.
- Validating the generated script.

> Your browser does not support embedded video. Watch the scenario recording demo (MP4).

> Video: Scenario Recording & AI Generation Demo

Demo 2 — CI/CD-Triggered Regression Testing

> This demonstration showcases TestPilot's integration with GitHub Actions.

> A code change is pushed to the separate SauceDemo application repository, triggering an automated CI/CD workflow.

> The workflow deploys the updated test application in the controlled demonstration environment, starts a TestPilot regression cycle, and retrieves its execution results.

> This illustrates how TestPilot can integrate regression testing into a continuous delivery workflow without requiring manual execution of every test campaign.

> Video: Automated Regression Testing with GitHub Actions

Demonstration Application — SauceDemo

> Both demonstrations use SauceDemo, a publicly available sample e-commerce application developed by Sauce Labs.

> SauceDemo was used exclusively as an external application under test and is not part of TestPilot's original development.

> Credits: SauceDemo — Sauce Labs (opens in a new tab)

Key Takeaways

> Developing TestPilot gave me hands-on experience in building a complete software engineering solution that combines full-stack development, agentic AI, browser automation, and DevOps.

> One of the most important lessons was that reliable AI-assisted automation requires more than generating code. It also depends on structured processing, validation, traceability, and human oversight.

> This project strengthened my ability to design interconnected services, solve integration challenges, and build practical AI-powered tools that support real software engineering workflows.

Animated content and alternate states

The project image card contains 7 manually switchable images. Previous/next buttons, slide indicators, keyboard navigation, and touch swipes select an image. Automatic rotation is disabled; the first image is initially selected.

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5	assets/img/portfolio/testpilot/record.png
6	assets/img/portfolio/testpilot/QA%20engineer%20dashboard.png
7	assets/img/portfolio/testpilot/qa%20manager%20dashboard.png

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The Solution	#the-solution	(empty)
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Watch the scenario recording demo (MP4)	assets/videos/testpilot-recorded-scenario.mp4	(empty)
SauceDemo — Sauce Labs (opens in a new tab)	https://www.saucedemo.com/	(empty)

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assets/img/portfolio/testpilot/BA%20dashboard.png	TestPilot Business Analyst dashboard showing scenario totals, test design progress and review tasks	(empty)
assets/img/portfolio/testpilot/record.png	TestPilot scenario recording form with project context, environment selection and scenario details	(empty)
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Image path	Exact alternate text	Role
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assets/img/portfolio/testpilot/AI-Powered%20Regression%20Testing%20Workflow.png	Illustration of manual test-script maintenance, campaign management and results analysis, with AI-generated scripts awaiting human review	(empty)

Demonstration videos

Demo 1 — From Scenario Recording to Automated Testing

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Inline playback	True
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10 · Digital Gallery

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Exact page content

Digital Gallery

> A personal collection of my digital drawings, made for the joy of creating.

> Spidey Digital drawing

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Main-content images

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11 · Contact

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Language	en
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Main section IDs	contact

Exact page content

Contact

> Have an opportunity, project idea, or collaboration in mind? Feel free to reach out.

Location

> Ariana, Tunisia

Email

> farahsayari.dev@gmail.com

Phone

> +216 28528390

Social Links

> GitHub | LinkedIn

Send me a message

> Your Name

> Your Email

> Subject

> Message

> Send Message

> You can also email me directly at farahsayari.dev@gmail.com.

Main-content links and downloads

Exact label / accessible label	Exact destination	Download filename
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+216 28528390	tel:+21628528390	(empty)
GitHub	https://github.com/FarahSayari	(empty)
LinkedIn	https://www.linkedin.com/in/farah-sayari-060745248	(empty)
farahsayari.dev@gmail.com	mailto:farahsayari.dev@gmail.com	(empty)

Embedded content

Exact title	Exact embed URL
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Contact form configuration and field copy

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Action	https://formspree.io/f/xgojgjbj

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Visibility	visible

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Visibility	visible

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Visibility	hidden

Website

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Required	no
Visibility	hidden

Conditional interface messages

> Loading

> Your message has been sent. Thank you!

> Please complete all fields with a valid email address.

> Message could not be sent. Please email me directly instead.

The browser validates required fields and the email format. On submission the inline script posts FormData to the configured action with Accept: application/json, shows a loading state, and disables the submit button. A successful response resets the form and shows the success message. An invalid form or unsuccessful request shows the corresponding message above. The _gotcha field is the hidden honeypot; the _subject field supplies the notification subject.

12 · Starter template

Page attribute	Exact value
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Keywords meta tag	(empty)
Main section IDs	starter-section

Retained template file. It is not linked from the main navigation or the current project list. Its template copy is reproduced below as part of the complete file inventory.

Exact page content

Starter Section

> Necessitatibus eius consequatur ex aliquid fuga eum quidem sint consectetur velit

> Use this page as a starter for your own custom pages.

Source inventory and current content differences

Current differences retained in this record

The document records the website as it exists. These items are observations for future editing; the website was not changed while preparing this reference.

Observation	Pages / current value
Header/footer email destinations	farahsayari.dev@gmail.com: about.html, certifications.html, contact.html, digital-gallery.html, index.html, portfolio-details.html, portfolio.html, pulmotrack-details.html, resume.html, services.html, starter-page.html, testpilot-details.html; farah.sayari@esprit.tn: about.html, certifications.html, digital-gallery.html, index.html, services.html
Project entries without descriptions or links	AI Agent Recruiter
Different project naming	The resume calls its career assistant Aura; the portfolio entry is titled AI Agent Recruiter. This reference does not assume that they are the same project.
Retained starter page	starter-page.html. Current brand: FARAH. Its exact content and links are recorded in its section.
Residual contact template backend	forms/contact.php remains present. Configured receiving address: contact@example.com. Current HTML form actions: https://formspree.io/f/xgojgjbj

Stylesheet and script references

Reference	Used by
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Favicons

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starter-page.html	apple-touch-icon	assets/img/apple-touch-icon.png

Downloadable documents

Local PDF file	Referenced by
assets/docs/Farah_Sayari_CV2026.pdf	resume.html
assets/docs/PulmoTrackPaper.pdf	pulmotrack-details.html

Each original linked PDF is preserved in Content_Source_Snapshot.zip. Its unmodified pages are appended to the PDF version of this reference; the Word/HTML/Markdown versions identify the documents rather than rewriting them. View and download labels and requested filenames appear in the page-specific link tables.

Source snapshot and verification

Content_Source_Snapshot.zip contains the captured HTML files, custom CSS and JavaScript, referenced local images and PDF documents, and the residual form/readme files. Source_Manifest.json records SHA-256 checksums. Content_Inventory.json contains the structured page inventory; Browser_Capture.json preserves the browser-parsed DOM and text. Vendor libraries are identified by reference rather than copied into the source archive.

The page transcripts are checked against every non-hidden text fragment inside each main element. Source hashes are checked before and after generation to confirm the website files remain unchanged. The generated PDF is checked for text coverage and readable page bounds.

Owner notes and future updates

These notes are editable in documentation/notes.json. They are not current website copy and are kept separately from the exact page transcripts.

No owner notes have been added yet.

FARAH SAYARI

AI & SOFTWARE ENGINEER

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SUMMARY

Recent Software Engineering graduate specializing in Data Science, with hands-on experience in Machine Learning, Deep Learning, agentic AI, and full-stack development. Skilled in developing intelligent, data-driven applications and seeking a full-time opportunity in AI, Data Science, or Software Engineering.

EDUCATION

Engineering Cycle in Computer Science – Data Science — **Sep 2023 – Nov 2026**
ESPRIT – Private Higher School of Engineering and Technology

Bachelor's Degree in Computer Systems Engineering – Embedded Systems & IoT — **Sep 2020 – Jun 2023**
Faculty of Sciences of Bizerte

SKILLS

Programming: Python, Java, JavaScript, R, C.
AI & Machine Learning: Scikit-learn, TensorFlow, Keras, PyTorch*, MLflow, Deep Learning, NLP, Recommendation Systems.
Generative AI & Agents: LLMs, RAG, LangChain, LangGraph, Qdrant, Ollama, Prompt Engineering.
Backend & Frontend: Spring Boot, FastAPI, Flask, Angular, React, REST APIs.
Data Engineering & Analytics: SQL, PostgreSQL, Hadoop, Talend, Data Warehousing, Power BI, Time Series Analysis.
DevOps & Tools: Git, Docker, GitHub Actions, Jenkins, Linux, Cypress, Kubernetes.

PROFESSIONAL EXPERIENCE

PFE Internship – VERMEG — **Feb – Aug 2026**
AI & Full-Stack Software Engineer Intern | TestPilot – AI-Assisted Test Automation Platform

- Designed and developed a full-stack test automation platform using **Angular, Spring Boot, FastAPI**, and **PostgreSQL**, enabling QA teams to manage, generate, and execute automated tests.
- Engineered a **hybrid agentic AI pipeline** using **Python, LangGraph,** and **LLMs** to transform recorded user interactions into structured test models and executable Cypress scripts.
- Developed a **Chrome extension (Manifest V3)** to capture browser interactions and implemented **LLM-assisted script repair** with validation and controlled retries.
- Integrated **Cypress** test execution, automated reporting, **GitHub Actions CI/CD**, and **Docker** to support continuous testing and deployment.

Summer Internship – Esprit-Tech — **Jul – Aug 2025**
Data Scientist | Clustering Algorithms for creation of heterogeneous student groups

- Implemented and compared multiple **clustering algorithms**, achieving a 0.72 Silhouette Score and building an interactive **Streamlit** dashboard to visualize and evaluate results.
- Developed an automated group formation pipeline integrated into the backend, supporting teachers in forming balanced and diverse student teams.

PROJECTS

LollyBooks – Personalized Digital Library & AI Reading Companion — **Oct - Present**
Co-Founder & Software Engineer | Ongoing Project

- Co-designed and developed an **interactive digital library** featuring customizable bookshelves, reading analytics, and goal tracking.
- Integrated the **Open Library REST API** and implemented **Goodreads CSV data processing** to automate book metadata retrieval and reading history imports.
- Prototyped a **personalized recommendation system** based on reading preferences and mood, and designed an **LLM-powered reading companion for conversational book discovery.**

Aura – Agentic AI Career Assistant — **Oct - Present**
AI & Software Engineer | Ongoing Project

- Redesigning a local, **agentic AI assistant** to automate job discovery, identify relevant opportunities, and match job descriptions with personal skills and experience.
- Developing an **LLM-powered application assistant** to generate tailored cover letters, analyze CV compatibility, and recommend targeted resume improvements.
- Adapting **RAG, semantic search**, and **agent orchestration** using **LangChain, LangGraph,** and **Qdrant** to support personalized job search and application workflows.

ADDITIONAL INFORMATION

- **Languages:** Arabic (Native), French (C2), English (C1).
- **Affiliations:** IEEE Computational Intelligence Society (CIS) Chapter, ESPRIT (2025).
- **Competitions & Awards:**
  - Bal des Projets – ESPRIT (2024 & 2025): Academic projects pre-selected for both editions.
  - IEEEXtreme 17.0 & 18.0 (2024–2025): Participant.
  - University English Competition (2024): 3rd Place.
- **Interests:** Reading & Literature, Digital Art & Illustration.

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INFORMS JOURNAL ON DATA SCIENCE

Scientific Paper – PulmoTrack Project

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PulmoTrack: A Deep Learning Solution for Pulmonary Nodule Segmentation, Detection, and Classification on CT and Chest X-ray Scans

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Abstract.

Background: Early detection and characterization of pulmonary nodules are critical for improving lung cancer prognosis. However, manual analysis of chest CT scans and X-rays is time-consuming and prone to variability. Deep learning (DL) methods offer the potential for automating and standardizing this process across imaging modalities.

Objective: We propose PulmoTrack, a multi-task, multi-modal deep learning framework for automated pulmonary nodule *segmentation*, *detection*, and *classification* (malignant vs. benign) on CT scans and multiclassification on chest X-ray images.

Methods: PulmoTrack was evaluated on two large public datasets: **LUNA16** (low-dose thoracic CT scans) and **NIH ChestXray** (posterior-anterior chest radiographs). For CT-based analysis, we developed a 3D U-Net for segmentation, a 3D encoder-based CNN for nodule detection, and a custom 3D convolutional classifier (Improved3DCNN) for malignancy classification. For chest X-rays, we designed 2D CNN architectures tailored to each task. The models were trained and validated using standardized preprocessing pipelines and data augmentation strategies.

Results: Across both datasets, PulmoTrack achieved high performance in all three tasks. On LUNA16, we report a Dice score of *0.95* for segmentation, a detection sensitivity of *92%*, and an AUC of *0.91* for malignancy classification. On NIH ChestXray, our 2D models achieved an AUC of *90%* in nodule multi-classification and strong visual performance in detection and segmentation.

Conclusion: PulmoTrack demonstrates the feasibility and effectiveness of a unified deep learning approach for pulmonary nodule analysis on CT and X-ray modalities. The system provides fast, accurate, and interpretable predictions, paving the way for more scalable and robust computer-aided diagnosis tools in clinical practice.

Keywords: Pulmonary nodules, Lung cancer, Deep learning, CT imaging, Chest X-ray, LUNA16, NIH ChestXray, Nodule segmentation, Nodule detection, Nodule classification, 3D CNN, YOLO V8, 3D Unet, Medical image analysis.

1. Introduction

Lung cancer remains the leading cause of cancer-related deaths globally, accounting for more than 1.8 million deaths annually. Early detection of pulmonary nodules, which are small masses in the lungs that may indicate early-stage cancer, is critical to improving patient outcomes. However, the manual identification and interpretation of these nodules in chest computed tomography (CT) and chest X-ray images is a time-consuming and error-prone process, especially given the high volume of imaging data in clinical practice.

Chest CT scans provide high-resolution volumetric data, making them the gold standard for pulmonary nodule analysis. Nonetheless, the sheer number of slices per scan often leads to reader fatigue and inter-observer variability among radiologists. Similarly, chest X-rays are widely used due to their low cost and accessibility, but they offer lower sensitivity and may miss subtle or overlapping nodules. In both modalities, precise and early diagnosis remains a challenge, especially when distinguishing benign from malignant nodules.

In recent years, deep learning (DL) has demonstrated significant promise in medical imaging tasks such as detection, segmentation, and classification. Convolutional Neural Networks (CNNs), in particular, have shown the ability to automatically extract meaningful features from raw medical images, reducing dependence on manual annotations and handcrafted features. However, most existing DL-based approaches address only a single task (e.g., detection or classification) or focus exclusively on a single imaging modality.

To address these limitations, we present **PulmoTrack**, a unified, multi-task deep learning framework designed to perform three key diagnostic tasks — *nodule segmentation*, *nodule detection*, and *nodule classification (malignant vs. benign)* — across two imaging modalities: CT scans and chest X-rays. Our system is built and evaluated using two benchmark datasets: **LUNA16** for CT data, and **NIH ChestXray** for chest radiographs. PulmoTrack integrates 3D and 2D CNN architectures tailored to the spatial characteristics of each modality, and is optimized for both accuracy and computational efficiency.

By unifying multiple diagnostic tasks across both CT and X-ray data, PulmoTrack represents a scalable and clinically relevant step toward comprehensive computer-aided diagnosis (CAD) systems for lung cancer screening and triage.

2. Dataset : LUNA16 (CT Scans)

The LUNA16 dataset (1) is a curated subset of the LIDC-IDRI dataset, containing 888 low-dose thoracic CT scans with annotated pulmonary nodules. Each nodule is labeled by up to four expert radiologists, and nodules with

a diameter of ≥ 3 mm are used for training and evaluation. For our purposes, we used the provided candidate coordinates for detection, voxel-wise masks for segmentation, and malignancy scores for classification. All volumes were resampled to isotropic resolution ($1 \times 1 \times 1$ mm) and cropped into $64 \times 64 \times 64$ patches centered around candidate nodules.

2.1. Data Preprocessing

For Luna16, preprocessing included resampling, intensity normalization, contrast enhancement, and data augmentation. CT volumes were augmented using random rotations, flips, and elastic deformations, while X-ray images were augmented via random cropping, horizontal flipping, and brightness/contrast adjustments.

2.2. Model Architectures

2.2.1. Nodule Segmentation For CT-based segmentation, we implemented a 3D U-Net (2) architecture composed of encoder-decoder layers with skip connections. Each convolutional block consisted of 3D convolutions, batch normalization.

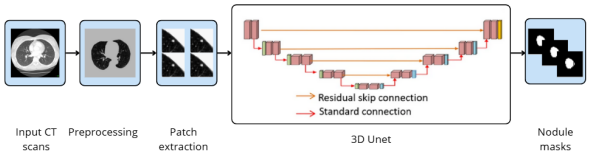


Figure 1 **3D-Unet Architecture For Nodule Segmentation.**

2.2.2. Nodule Detection For detection, a 3D CNN classifier scans sub-volumes of the CT images and outputs bounding boxes with likelihood scores for nodule presence.

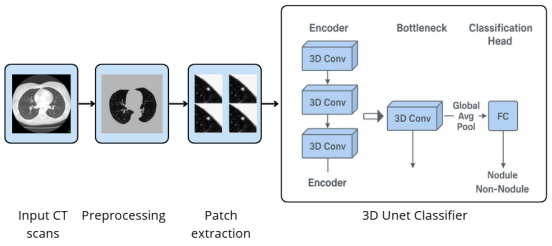


Figure 2 **3D-Unet Classifier Architecture For Nodule Detection**

2.2.3. Nodule Classification The classification module distinguishes between benign and malignant nodules using an improved 3D CNN model trained with binary cross-entropy and enhanced with dropout and batch normalization layers.

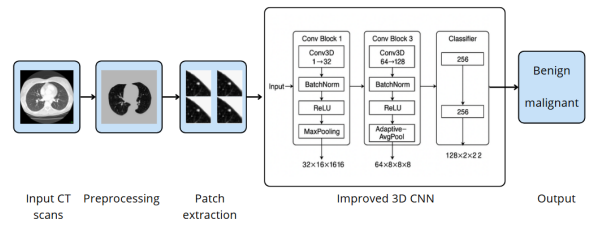


Figure 3 **Improved3DCNN Architecture For Nodule Classification**

2.3. Training Procedure

All models were implemented in PyTorch and trained on NVIDIA GPUs. Binary cross-entropy was used for classification and detection, and a combination of Dice loss and cross-entropy was used for segmentation. Optimization was done using Adam with an initial learning rate of 1e-4. Early stopping and model checkpointing were employed based on validation loss.

2.4. Results

2.4.1. Evaluation Metrics We evaluated the models using the following metrics:

• **Segmentation:** Dice Similarity Coefficient (DSC), Jaccard Index (IoU)

• **Detection:** Sensitivity, Specificity, False Positives per Scan (FPs), Free-Response Receiver Operating Characteristic (FROC)

• **Classification:** Area Under the Receiver Operating Characteristic Curve (AUC), Accuracy, Precision, Recall, F1-score

2.4.2. Segmentation Our 3D U-Net achieved a mean Dice score of **0.95** on the test set. The model was able to capture nodule boundaries across a wide range of sizes accurately. However, it still struggles to precisely segment very tiny nodules (typically < 4 mm in diameter), which are more prone to boundary ambiguity and are often missed due to their low contrast and limited voxel footprint.

Detection : The encoder-based 3D CNN classifier reached a sensitivity of **92%** at 1.5 FPs per scan. The FROC curve (see Figure 3) demonstrates high discriminative ability across varying thresholds.

2.4.3. Classification The Improved3DCNN model achieved an AUC of **0.91**, an accuracy of **89%**, and an F1-score of **0.88**, showing strong capability to differentiate benign from malignant nodules.

Conclusion As summarized in Table 1, our PulmoTrack pipeline achieved strong performance across all three diagnostic tasks on the LUNA16 dataset. Notably, when comparing our results with those reported by Gao et al. (3), who conducted a systematic review of deep learning methods for pulmonary nodule detection and segmentation in CT scans, our approach demonstrates superior accuracy and segmentation quality. Specifically, our 3D U-Net achieved a Dice score of 0.95, which surpasses the typical performance range (0.80–0.92) observed across multiple studies reviewed in their paper. Similarly, our detection sensitivity of 92% at 1.5 false positives per scan and classification AUC of 0.91 are competitive with or better than state-of-the-art benchmarks highlighted in that review. These findings validate the robustness and clinical potential of the PulmoTrack framework.

3. Dataset: NIH ChestXray

The NIH ChestXray dataset (4), publicly available on Kaggle, contains 112,120 frontal chest

Table 1 Performance summary on LUNA16 dataset.

Task	Metric	Score
Segmentation	Dice Score	95%
Detection	Sensitivity @ 1.5FP	92%
Classification	AUC	91%

X-ray images from 30,805 patients, each annotated with one or more of 14 disease labels, including Nodule, Mass, Effusion, and Cardiomegaly. It includes two main metadata files: `Data_Entry_2017.csv`, with image filenames, multi-label annotations, patient info, and view position; and `BBox_List_2017.csv`, containing bounding box coordinates for 8 diseases. Images may have multiple labels or be marked as "No Finding," which occurs in 60,361 cases. The dataset is imbalanced; for example, Hernia appears in only 227 images. This study focuses on the Nodule class (6,331 samples), which influenced data filtering and design choices. Images are in PNG format with varying resolutions (e.g., 2048×2500 pixels). The dataset is widely used for multi-label classification, weakly supervised learning, and localization tasks.

3.1. Pipeline A – Segmentation and Mask Generation

3.1.1. Motivation A significant limitation of the NIH ChestXray dataset is the absence of pixel-wise segmentation masks, which restricts its application in fully supervised segmentation tasks. To overcome this, a segmentation pipeline was developed to generate lung region masks, facilitating spatial learning and localization.

3.1.2. Annotated Data and Auxiliary Sources To train a supervised segmentation model, three external datasets providing ground-truth lung masks were utilized:

- **JSRT Dataset :** Comprising 247 chest radiographs with and without nodules, each accompanied by expert-annotated lung segmentation masks.
- **Montgomery County X-ray Set :** Consists of 138 posterior-anterior chest X-rays (normal and tuberculosis cases), all manually segmented by clinical experts.
- **NIH Subset with Expert Annotations :** A subset of 100 NIH images, externally sourced, includes radiologist-annotated lung masks. This subset provides a direct correspondence between NIH images and high-quality segmentation labels.

These datasets were used to train a general-purpose lung segmentation model, which was subsequently applied to the broader NIH dataset to generate synthetic lung masks for use in downstream tasks.

3.1.3. Preprocessing and Augmentation for Segmentation To ensure consistent input formatting, all chest X-ray images and corresponding segmentation masks underwent a standardized preprocessing pipeline. Images were converted to grayscale, resized to 224 ×

224 pixels, enhanced using histogram equalization to improve local contrast, and normalized to the $[0, 1]$ range for training stability. Masks were resized to 224×224 pixels using nearest-neighbor interpolation to preserve label integrity and normalized to the $[0, 1]$ range, as they are treated as floating-point tensors in the model input.

3.1.4. Dataset Splitting The aggregated dataset was divided into three subsets to support robust model development and evaluation: 80% for the training set, 10% for the validation set, and 10% for the test set.

This stratified split ensured that the model had sufficient data for learning while retaining dedicated subsets for unbiased validation and final performance testing.

3.1.5. Model Architecture The segmentation model was built using a U-Net architecture implemented via the `segmentation_models_pytorch` (SMP) library, selected for its modularity and support for pre-trained backbones. The encoder was a ResNet34 pretrained on ImageNet, with input channels set to 3 (RGB format) and output channels set to 1 (binary segmentation mask). This architecture leverages deep residual features from the encoder while enabling precise spatial reconstruction through the decoder, making it well-suited for medical image segmentation.

3.1.6. Loss Function To address class imbalance and improve overlap quality in segmentation, a custom loss function was designed—Dice + Binary Cross Entropy (DiceBCELoss). Binary Cross Entropy (BCE) captures pixel-wise classification accuracy, while the Dice coefficient emphasizes accurate region overlap, which is particularly important for small anatomical structures. This combined loss enhances the model's capacity to correctly learn both the presence and shape of the lung regions.

3.1.7. Evaluation Metrics Model performance was quantitatively assessed using the following evaluation metrics:

- Accuracy
- Sensitivity (Recall)
- Specificity
- Dice Coefficient
- Jaccard Index (Intersection over Union, IoU)

These metrics collectively provide a detailed evaluation of segmentation quality, addressing both pixel-level correctness and region-level overlap.

3.1.8. Model Performance and Results The final model was selected based on the highest validation Dice coefficient, which occurred at Epoch 10. This checkpoint demonstrated strong generalization performance across multiple segmentation metrics.

These results indicate consistent performance across the training and validation datasets, with

Table 2 Performance Summary at Epoch 10

Metric	Training	Validation
Dice Coefficient	0.7727	**0.8521** (Best)
Jaccard Index (IoU)	0.6296	0.7423
Accuracy	0.8389	0.9058
Sensitivity (Recall)	0.9985	0.9985
Specificity	0.7786	0.8712
Combined Loss	-1.3446	-1.6483

particularly strong outcomes in sensitivity and segmentation overlap metrics, highlighting the model's robustness and generalization capability.

3.1.9. Observations and Insights The model demonstrated excellent foreground detection, achieving a sensitivity of 99.85%, which indicates highly reliable identification of lung regions with minimal false negatives. A high Dice coefficient (0.8521) and Jaccard Index (0.7423) reflect strong performance in terms of segmentation overlap and class balance, suggesting an effective balance between precision and recall. The validation accuracy of 90.58% highlights the model's strong generalization capability. Furthermore, the steady decrease in the combined loss (Dice + BCE) during training confirms stable convergence and effective learning behavior.

Conclusion Epoch 10 was selected as the final model due to its superior validation performance. It achieved high-quality segmentation with near-perfect sensitivity, balanced accuracy, and strong generalization. These results confirm the effectiveness of the proposed pipeline for downstream spatial or region-based tasks.

3.2. Pipeline B – Pathology Classification and Analysis

3.2.1. Preprocessing for Classification Before model development, the NIH ChestXray metadata was reviewed, and columns with predominantly missing values were removed to reduce noise and improve reliability. For image preprocessing, the same standardized steps from Pipeline A were applied to ensure consistency: grayscale conversion, resizing to 224×224 pixels, histogram equalization, and normalization to the [0, 1] range. This ensured clean, consistent inputs across both metadata and image modalities, supporting effective classification model development.

3.2.2. NIH ChestXray Labeling To generate spatially-aware supervision, the segmentation model from Pipeline A was applied to the entire NIH ChestXray dataset, producing lung region masks for each image. These binary masks highlighted pulmonary fields and served as intermediate annotations rather than direct inputs for classification. This approach enriched the dataset with pseudo-segmented regions, facilitating a weakly-supervised classification framework.

3.2.3. Postprocessing via PS-KDE To enhance the quality of the predicted masks and

reduce noise, a postprocessing step was introduced using **Probability Smoothing with Kernel Density Estimation (PS-KDE)**.

This method refines the raw segmentation masks by:

- Smoothing noisy pixel-level predictions
- Reinforcing consistent structural regions (e.g., lung contours)
- Eliminating spurious artifacts from the segmentation output
- Reducing image noise introduced by medical equipment or acquisition artifacts

The PS-KDE postprocessing was applied jointly on the original preprocessed images and their predicted lung masks, resulting in a new set of enhanced segmented images. These images maintain high anatomical fidelity while preserving essential pathological cues, and they serve as the final inputs for the classification model in **Pipeline B**.

This postprocessing step ensures that only high-confidence lung regions contribute to the learning process, improving model robustness, image quality, and interoperability.

3.2.4. Training Results and Model Performance The final model in **Pipeline A** was trained for up to 50 epochs, with early stopping triggered at epoch 15 due to stagnating validation performance. The best-performing checkpoint achieved the following metrics:

- **Training Loss:** 0.8479 **Validation Loss:** 2.1292
- **Training AUC:** 0.9024 **Validation AUC:** 0.7983
- **Training F1 Score:** 0.3056 **Validation F1 Score:** 0.2342
- **Training Accuracy:** 0.7693 **Validation Accuracy:** 0.7720
- **Training Exact Match Accuracy:** 0.0190 **Validation Exact Match Accuracy:** 0.0213

The model showed strong discriminative capability, as reflected in the high AUC scores on both training and validation sets. However, F1 scores and Exact Match Accuracy remained modest—primarily due to the challenges of multi-label classification and significant class imbalance.

Metric interpretation:

- **Accuracy** indicates the average proportion of correct labels per sample. It allows partial correctness (e.g., predicting 3 out of 5 labels yields 60% accuracy).
- **Exact Match Accuracy** is stricter: it requires an exact match between predicted and ground truth labels. Any deviation leads to a failed prediction, making this metric particularly difficult in the presence of label co-occurrence and imbalance.

Compared to earlier iterations that suffered from overfitting (high training scores but poor validation performance), this version showed improved generalization. Key improvements included:

- Targeted data augmentation strategies

- Use of an adaptive optimizer
- Class-frequency-based loss weighting to mitigate imbalance

Together, these enhancements yielded a more robust and stable model, establishing Pipeline A as a strong baseline and paving the way for the segmentation-aware classification in **Pipeline B**.

4. Discussion

PulmoTrack demonstrates strong performance across key tasks in lung imaging analysis using both CT and chest X-ray modalities. On the LUNA16 CT dataset, our unified 3D pipeline achieved competitive results in segmentation, detection, and malignancy classification of lung nodules. For chest X-ray analysis, the integration of PS-KDE preprocessing and a ResNet34-based U-Net led to high-precision segmentation of anatomical structures across JSRT, Montgomery, and NIH datasets. In addition, we successfully developed a multi-class classification model to detect multiple thoracic diseases from X-ray images, achieving strong performance aligned with state-of-the-art benchmarks. These results confirm the robustness of our approach. While each modality was handled through a distinct pipeline, both demonstrated generalization and diagnostic potential. Future integration of both modalities could further enhance clinical applicability.

5. Discussion

PulmoTrack demonstrates robust performance across key diagnostic tasks in lung imaging using both CT and chest X-ray modalities. On the LUNA16 dataset, our unified 3D pipeline achieved strong results in segmentation, detection, and malignancy classification of pulmonary nodules. For chest X-ray analysis, the integration of PS-KDE postprocessing and a ResNet34-based U-Net enabled high-precision segmentation of anatomical structures across JSRT, Montgomery, and NIH datasets. Furthermore, our multi-label classification model successfully detected a wide range of thoracic diseases from X-ray images, yielding performance aligned with or exceeding recent benchmarks.

These outcomes highlight the reliability and generalization capability of our dual-pipeline framework. While CT and X-ray data were processed through distinct architectures, both pipelines demonstrated clinical relevance and scalability. Notably, our approach achieved performance surpassing models reviewed in Gao et al. (3), especially on LUNA16.

Nonetheless, certain limitations remain—such as the challenge of segmenting very small nodules and the weak supervision inherent in some X-ray datasets. Future improvements may involve attention-based mechanisms, class-balancing strategies, and label refinement through semi-supervised learning. Additionally, integrating information

from both modalities through multimodal fusion could enhance diagnostic precision and adaptability to diverse clinical scenarios.

6. Conclusion

PulmoTrack introduces a dual-modality deep learning framework for comprehensive pulmonary analysis using CT and chest X-ray imaging. On the CT side, leveraging the LUNA16 dataset, our system performed accurate lung nodule segmentation, detection, and malignancy classification using 3D architectures. On the X-ray side, the combination of PS-KDE postprocessing and a ResNet34-based U-Net enabled high-fidelity anatomical segmentation, while a multi-label classification model effectively identified multiple thoracic pathologies.

Our segmentation models demonstrated strong generalization across multiple chest X-ray datasets (JSRT, Montgomery, NIH), often with minimal retraining. Importantly, the overall framework outperformed prior benchmarks in key evaluation tasks, particularly on the LUNA16 dataset, reinforcing its potential for AI-assisted pulmonary diagnosis.

In future work, we aim to explore multimodal fusion techniques to jointly exploit CT and X-ray data, while scaling evaluation to larger and more diverse patient cohorts. We also envision improvements in explainability and clinical integration to support real-world deployment in radiology workflows.

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